

Finding a ratio

Large numbers can also be written as ratios. For example, to find the ratio between 1 hour and 20 minutes, convert them into the same unit, then cancel these numbers by finding the highest number that divides into both.

minutes are the smaller unit
1 hour is the same as 60 minutes, so convert
this is the symbol for ratio

$$20 \text{ mins, } 60 \text{ mins} \rightarrow 20 : 60$$

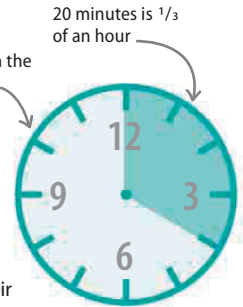
Convert one of the quantities so that both have the same units. In this example use minutes.

Write as a ratio by inserting a colon between the two quantities.

ratios show information in the same way as fractions do
 $60 \div 20 = 3$
 $20 \div 20 = 1$

$$1 : 3$$

Cancel the units to their lowest terms. Here both sides divide exactly by 20 to give the ratio 1 : 3.

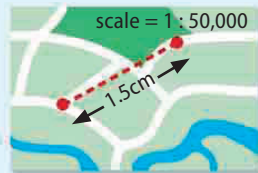


Working with ratios

Ratios can represent real values. In a scale, the small number of the ratio is the value on the scale model, and the larger is the real value it represents.

Scaling down

1 : 50,000 cm is used as the scale on a map. Find out what a distance of 1.5 cm represents on this map.



scale shows what real distance is represented by each distance on the map

distance on map
scale on map
actual distance represented by the map

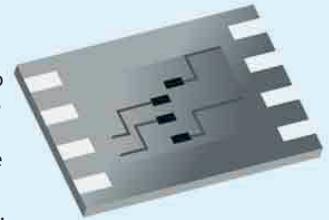
$$1.5 \text{ cm} \times 50,000 = 75,000 \text{ cm}$$

$$= 750 \text{ m}$$

the answer is converted into a more suitable unit—there are 100 cm in a meter

Scaling up

The plan of a microchip has the scale 40 : 1. The length of the plan is 18 cm. The scale can be used to find the length of the actual microchip.



length of plan
divide by scale to find actual size
actual length of microchip

$$18 \text{ cm} \div 40 = 0.45 \text{ cm}$$

Comparing ratios

Converting ratios into fractions allows their size to be compared. To compare the ratios 4 : 5 and 1 : 2, write them as fractions with the same denominator.

fraction that represents ratio 1 : 2
fraction that represents ratio 4 : 5

$$1 : 2 = \frac{1}{2}$$

and

$$4 : 5 = \frac{4}{5}$$

2 × 5 is 10, the common denominator
5 × 2 is 10, the common denominator

$$\frac{1}{2} = \frac{5}{10}$$

$$\frac{4}{5} = \frac{8}{10}$$

First write each ratio as a fraction, placing the smaller quantity in each above the larger quantity.

Convert the fractions so that they both have the same denominator, by multiplying the first fraction by 5 and the second by 2.

compare the numerators

$$\frac{5}{10} \text{ is smaller than } \frac{8}{10}$$

so

$$1 : 2 \text{ is smaller than } 4 : 5$$

Because the fractions now share a denominator, their sizes can be compared, making it clear which ratio is bigger.